

What is bronchiolitis?

It is a viral lower respiratory tract infection. It generally affects infants (under 12 months of age), but can affect children up to 2 years of age. From 1-2 years of age, there is some overlap with viral-induced wheeze and some risk of asthma.

Primary pathophysiology: air trapping, airway edema. Crackles are commonly heard in bronchiolitis.

[CLICK HERE TO JUMP TO MANAGEMENT](#)

History

Knowledge of the natural course can provide caregivers with appropriate anticipatory guidance (e.g. timing of when the disease is expected to be peaking)

Disease severity peaks usually on day 2-3 of illness.

For Respiratory Syncytial Virus, the peak can occur later - at day 5-7.

Total duration: usually 7-10 days, up to 2 weeks

Cough: can persist for several weeks

Patients who are higher risk of severe disease

- Infants less than 3 months of age
- Ex-premature infants
- Chronic heart disease (especially those with pulmonary hypertension)
- Chronic pulmonary disease
- Neurological conditions (especially those with impaired cough reflex)
- Maternal smoking
- Immunodeficiency

Examination:

Typical features include:

- Bilateral crackles +/- wheezing
- Increased work of breathing
- Respiratory distress (mild vs moderate vs severe)

Complications:

- Poor feeding
- Hypoxia
- Apneic episodes (in younger infants, consider pertussis as a differential)
- Respiratory failure

Severity Assessment

Valid only in otherwise well infants; Severity assessment also varies depending on comorbidities. Always remember to consider other differentials that may masquerade as bronchiolitis (e.g. congestive cardiac failure).

	Mild	Moderate	Severe
Mental state	Normal	At most mildly irritable	Irritable / Drowsy
Respiratory rate ⁺⁺ and distress	Normal for age, no distress	Increased RR	Marked increase in RR, or paradoxically relatively bradypneic
Use of accessory muscles	Nil	Present	Present
Feeding	Normal	May be impaired	Significantly impaired / Unsafe to feed orally
Oxygen requirement	SpO ₂ $\geq$ 92% without support	SpO ₂ $\geq$ 92% without support	Hypoxic, SpO ₂ $<$ 92%
Apnea	Nil	May have brief episodes [#]	Frequent or prolonged
Cyanosis	Nil	Nil	May be present

- If parental anxiety is significant, and/or concerns that caregiver ability is impaired
- Admit the patient
- Consider admitting the high risk young infant with bronchiolitis (less than 3 months of age)

⁺⁺ Count respiratory rate over 1 min to account for periodic breathing

[#] If there is ANY history of apnea - the child requires admission for inpatient monitoring and continuous pulse oximetry (infant apnea = cessation of breathing $>$ 20 sec)

[\(Click here to jump to Management\)](#)

Differentials

1. Pulmonary
 - a. Asthma (especially if there is a personal or family history of atopy and/or interval symptoms)
 - b. Pneumonia (especially when there are persistent unilateral signs)
 - c. Congenital lung disease (bronchiolitis can unmask a structural anomaly e.g. vascular ring)
 - d. Cystic fibrosis
 - e. Inhaled foreign body
2. Cardiac
 - a. Congenital heart disease / Congestive cardiac failure
3. Sepsis
4. Severe metabolic acidosis from other causes

Investigations

Usually NO INVESTIGATIONS are needed for typical bronchiolitis.

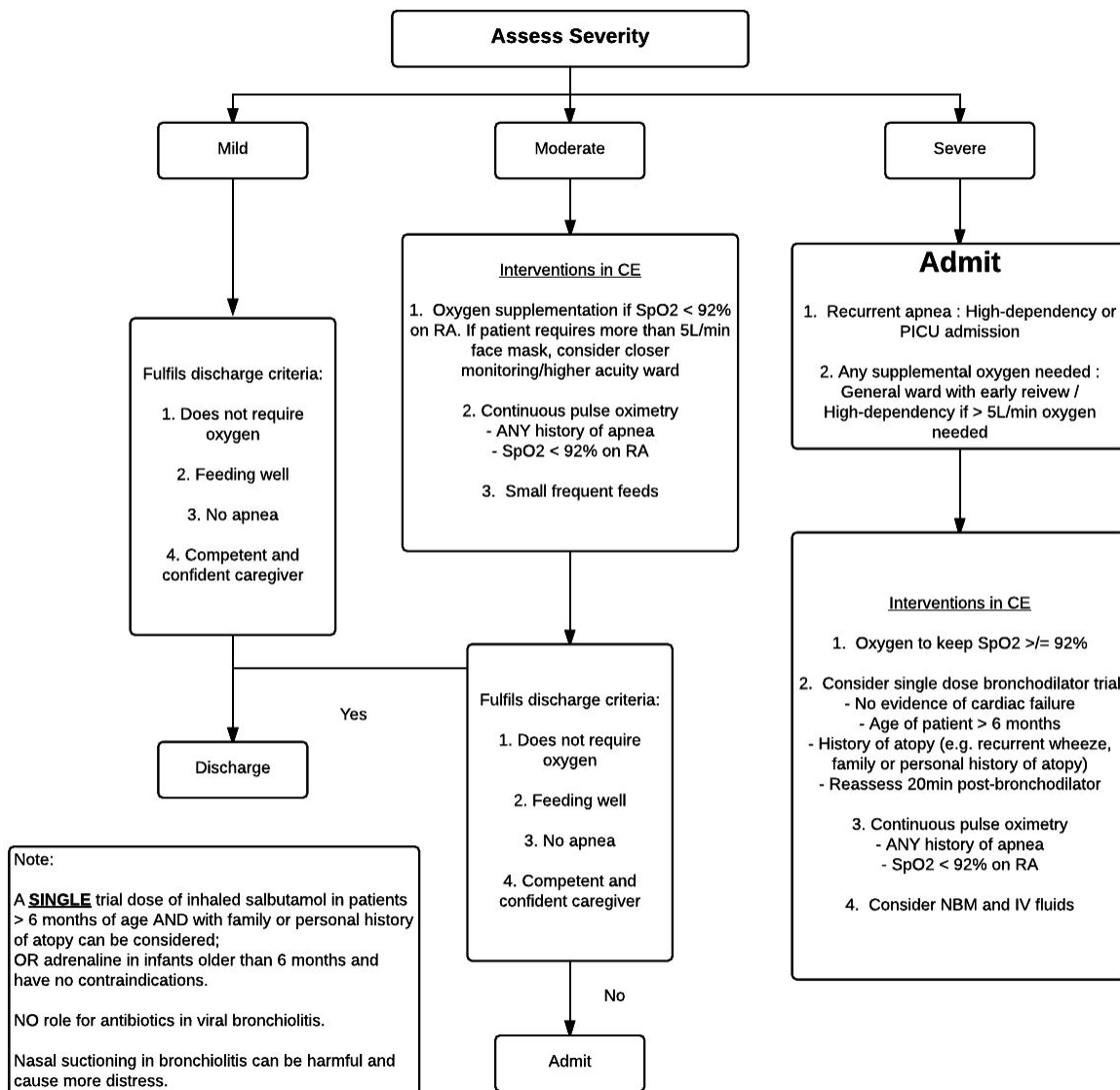
On a case-by-case basis, consider:

- Capillary blood gas (CBG) or Arterial blood gas (ABG) if severe disease / entering respiratory failure
- Nasopharyngeal aspirate if severe disease / at risk infant
- Chest X-ray if unilateral signs / diagnostic uncertainty / cardiac disease (e.g. cardiac murmur heard, concerns for congestive cardiac failure)
- Limited septic screen if clinically warranted / febrile infant < 3 months of age per our young infantile pyrexia guideline

Acute Management in Children's Emergency (CE)

Mainstay of therapy is **SUPPORTIVE** and possibly **CLOSE MONITORING** for deterioration. ([Click here to jump to severity assessment](#))

Management workflow in CE



The evidence to date:

1. Inhaled Salbutamol : still shows no benefit for typical bronchiolitis. A trial of salbutamol therapy can be considered as there is sometimes an element of bronchoconstriction.
2. Nebulized adrenaline : may improve symptoms within first 24 hours of clinical contact but makes no difference to eventual hospitalization rates / length of inpatient stay. (*Observe in CE at least 4 hours after nebulized adrenaline*)
3. Corticosteroids : No benefit
4. Chest physiotherapy : No benefit
5. Nasal or airway suctioning : No benefit and may cause harm. Increases distress in the patient.
6. Antibiotics : No role in viral bronchiolitis.
7. Hypertonic saline : No benefit in CE (may have role in reducing length of stay for inpatients)
8. High flow nasal canula (HFNC) : no role in emergency department, may have a role in admitted patients in reducing rate of PICU admission
9. Ribavirin : No role in CE, require inpatient respiratory team assessment

References:

1. Kepreotes E, Whitehead B, Attia J, et al. **High-flow warm humidified oxygen versus standard low-flow nasal cannula oxygen for moderate bronchiolitis (HFWHO RCT): an open, phase 4, randomised controlled trial.** Lancet. 2017 Feb 1. pii: S0140-6736(17)30061-2. doi: 10.1016/S0140-6736(17)30061-2. (Original) PMID: 28161016
2. Diagnosis and management of bronchiolitis. American Academy of Pediatrics Subcommittee on Diagnosis and Management of Bronchiolitis. Pediatrics. 2006;118(4):1774
3. Scottish Intercollegiate Guidelines Network. Bronchiolitis in children. A national clinical guideline. 2006
4. Accuracy of pulse oximetry in children. Accuracy of pulse oximetry in children. Pediatrics. 2014 Jan;133(1):22-9. Epub 2013 Dec 16.
5. Cochrane Database Syst Rev. 2012;2:CD004873. Roquéi Figuls M, Giné-Garriga M, Granados Rugeles C, Perrotta C Chest physiotherapy for acute bronchiolitis in paediatric patients between 0 and 24 months old.
6. Hartling L, et al. Epinephrine for bronchiolitis; Cochrane Database Syst Review 2010
7. Skjerven HO et al, Racemic adrenaline and inhalation strategies in acute bronchiolitis. NEJM 2013 368:2286
8. [Mandelberg A, Tal G, Witzling M, et al. Nebulized 3% hypertonic saline solution treatment in hospitalized infants with viral bronchiolitis. Chest 2003; 123:481.](#)
9. [Tal G, Cesar K, Oron A, et al. Hypertonic saline/epinephrine treatment in hospitalized infants with viral bronchiolitis reduces hospitalization stay: 2 years experience. Isr Med Assoc J 2006; 8:169.](#)
10. [Luo Z, Liu E, Luo J, et al. Nebulized hypertonic saline/salbutamol solution treatment in hospitalized children with mild to moderate bronchiolitis. Pediatr Int 2010; 52:199.](#)
11. [Kuzik BA, Al-Qadhi SA, Kent S, et al. Nebulized hypertonic saline in the treatment of viral bronchiolitis in infants. J Pediatr 2007; 151:266.](#)
12. [Luo Z, Fu Z, Liu E, et al. Nebulized hypertonic saline treatment in hospitalized children with moderate to severe viral bronchiolitis. Clin Microbiol Infect 2011; 17:1829.](#)
13. [Miraglia Del Giudice M, Saitta F, Leonardi S, et al. Effectiveness of nebulized hypertonic saline and epinephrine in hospitalized infants with bronchiolitis. Int J Immunopathol Pharmacol 2012; 25:485.](#)

14. [Sarrell EM, Tal G, Witzling M, et al. Nebulized 3% hypertonic saline solution treatment in ambulatory children with viral bronchiolitis decreases symptoms. Chest 2002; 122:2015.](#)
15. [Grewal S, Ali S, McConnell DW, et al. A randomized trial of nebulized 3% hypertonic saline with epinephrine in the treatment of acute bronchiolitis in the emergency department. Arch Pediatr Adolesc Med 2009; 163:1007.](#)
16. [Anil AB, Anil M, Saglam AB, et al. High volume normal saline alone is as effective as nebulized salbutamol-normal saline, epinephrine-normal saline, and 3% saline in mild bronchiolitis. Pediatr Pulmonol 2010; 45:41.](#)
17. [Al-Ansari K, Sakran M, Davidson BL, et al. Nebulized 5% or 3% hypertonic or 0.9% saline for treating acute bronchiolitis in infants. J Pediatr 2010; 157:630.](#)
18. [Jacobs JD, Foster M, Wan J, Pershad J. 7% hypertonic saline in acute bronchiolitis: a randomized controlled trial. Pediatrics 2014; 133:e8.](#)